Week 6: Introduction to Variational Inference
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What stood out to me in Blei, Kucukelbir, and McAuliffe's review was that inference becomes an optimization problem rather than a sampling problem. We choose a family of densities over the latent variables and search within it for the member closest to the posterior in KL divergence. The obstacle is that this divergence contains the log evidence, the quantity we could not compute in the first place, so we maximize the evidence lower bound instead. Because the ELBO equals the log evidence minus that divergence, maximizing it minimizes the divergence without ever evaluating the constant. I found the second form, $\mathbb{E}[\log p(x \mid z)] - \mathrm{KL}(q(z) \| p(z))$, the clearest way to see the tradeoff: fit to the observed data against staying close to the prior. It also makes week 3's stochastic optimization directly relevant, since the ELBO is one more objective to climb.

The cost of this convenience is where the review was most useful to me. The mean-field family treats the latent variables as mutually independent, each with its own factor, which makes coordinate ascent cheap but removes any ability to represent dependence among them. The authors are direct that variational inference generally underestimates posterior variance, a consequence of the objective rather than a fixable implementation detail. They also say plainly that the relative accuracy of variational inference and MCMC is still not understood. That contrast sharpens week 4 for me: MCMC pays in computation for samples that are asymptotically exact, while this pays in accuracy for speed, and the value of the ELBO at convergence does not report how large that error is.

Jordan, Ghahramani, Jaakkola, and Saul arrive at the same methods from an older direction. Their emphasis is convex duality: a concave function can be written as a minimum over a family of linear upper bounds, as with $\ln x \leq \lambda x - \ln \lambda - 1$, where each choice of the variational parameter λ is tight at one point. Fitting those parameters then turns an intractable graphical model into a simpler one where inference runs efficiently, and what comes back is a bound on the quantity of interest. Reading the two together, the question I am left with is how to check whether the independence assumption has distorted the particular posterior summary I care about, given that the bound itself will not tell me.

Readings: Blei et al. (2017), *Variational Inference: A Review for Statisticians*; Jordan et al. (1999), *An Introduction to Variational Methods for Graphical Models*.